Area, Surface Area & Volume

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

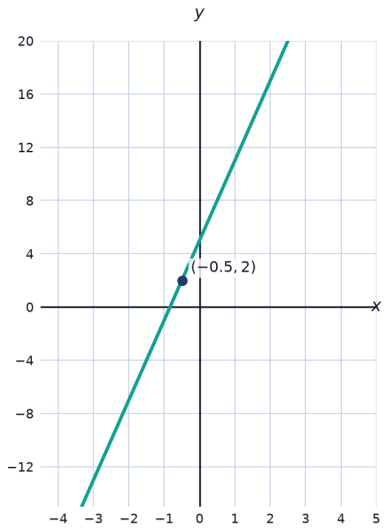
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$.

What is the arithmetic mean? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

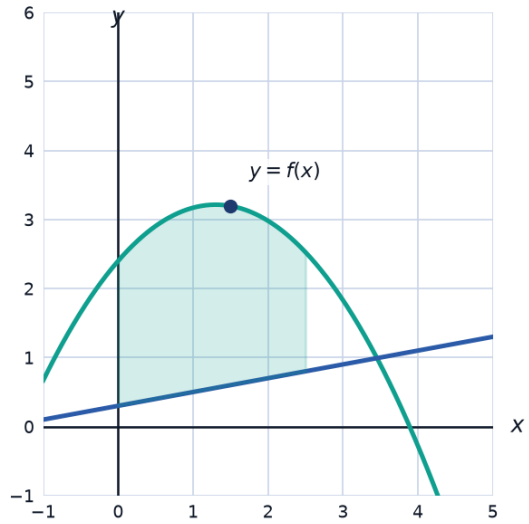
For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(3)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

Show your complete algebraic work.



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4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
A right circular cylinder has radius 3 cm and height 8 cm. What is its total surface area in terms of π?

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
A cuboid has a length of USD 8$cm, a width of USD 5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Find the area of a triangle with base 10 and height 6.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)
Find the area of a circle with diameter 10.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
A circle has a radius of 12 cm. What is the area of a sector formed by a central angle of 75°?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Find the volume of a rectangular prism $4 \times 3 \times 5$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Two similar cylinders have radii 4 cm and 6 cm. If the volume of the smaller is 160π cm³, find the volume of the larger.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Level 3 · AdvancedA school administrator randomly surveys 120 high school seniors at Oakwood High School. In the sample, 65% report planning to attend a four-year college, with a margin of error of $\pm 4\%$. Which of the following conclusions is most statistically sound? (A) Exactly 65% of all Oakwood High seniors will attend college. (B) It is plausible that between 61% and 69% of all Oakwood High seniors plan to attend a four-year college. (C) Between 61% and 69% of all high school seniors in the state plan to attend college. (D) To reduce the margin of error by half, the administrator should survey 240 seniors.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Level 3 · AdvancedUsing the student prep table (Studied: 54 passed, 6 failed; Did not study: 18 passed, 22 failed; Total passed: 72 out of 100), are the events 'passed' and 'studied' independent?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)
A line passes through $(1, 2)$ with slope 3. Find y when $x = 4$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line has slope 3 and passes through $(0, 2)$. Find y when $x = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Lines & Angles

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Triangles & Congruence

Practice 3.2: What is the domain of the function $f(x) = \frac{\sqrt{x+5}}{x-3}$?

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

For $f(x) = x^2$, find the average rate of change on $[1, 3]$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

Level 3 · AdvancedA quadratic function $f(x) = ax^2 + bx + c$ passes through the points $(-2, 11)$, $(0, 3)$, and $(2, 7)$. What is the value of $a + b + c$?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A population of bacteria in a petri dish starts with 250 organisms and quadruples every 5 hours. How many hours will it take for the population to reach 16,000?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Circles

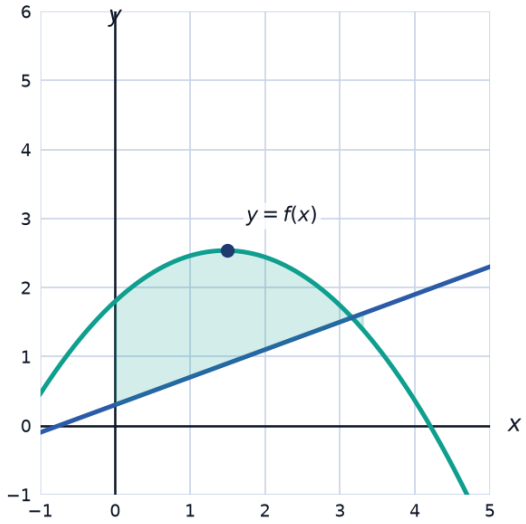
How many real solutions does $x^2 + 2x + 5 = 0$ have?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 88°

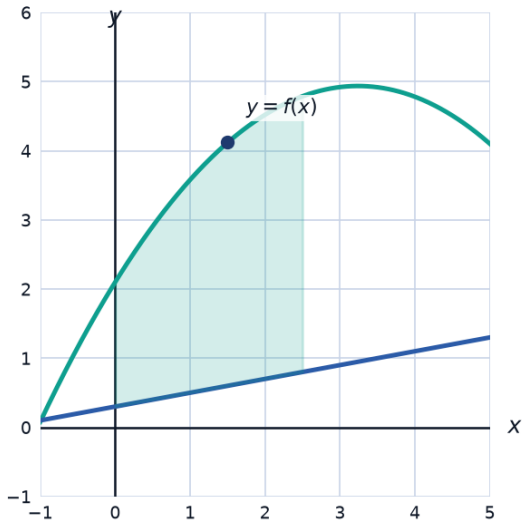
b. 98°

c. 80°

d. 133°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 87°

b. 97°

c. 79°

d. 132°

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70°

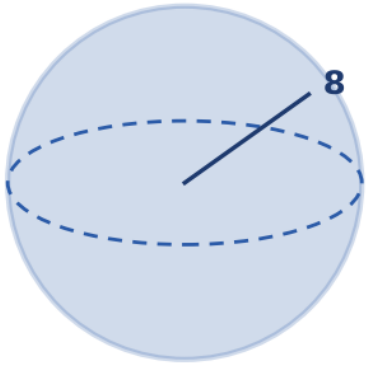
b. 80°

c. 62°

d. 129°

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A sphere has radius 8. Find its surface area in terms of π.



a. 128π

b. 262π

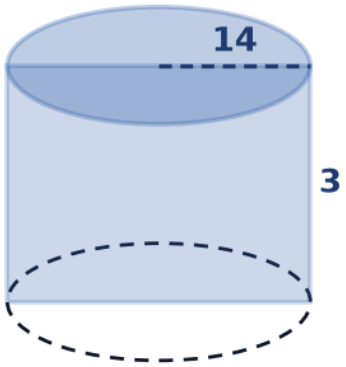
c. 256π

d. 64π

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A cylinder has radius 14 and height 3. Find its volume in terms of π.



a. 588π

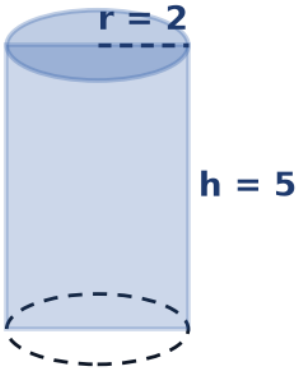
b. 42π

c. 1176π

d. 598π

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Volume of a cylinder with radius 2 and height 5 (in terms of π):



a. 10π

b. 25π

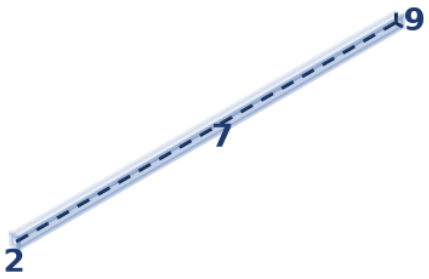
c. 40π

d. 20π

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A rectangular prism has length 2, width 7, and height 9. What is its volume?



a. 126

b. 136

c. 18

d. 190

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Area of a circle with radius 3 (in terms of π):

a. 12π

b. 7π

c. 18π

d. 9π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Surface area of a cube with side length 2:

a. 8

b. 16

c. 20

d. 24

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Area of a rectangle with length 3 and width 5:

a. 18

b. 8

c. 16

d. 15

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Two similar solids have corresponding side lengths in ratio $2:3$. Find the ratio of their volumes.

a. 4:9

b. 8:27

c. 6:9

d. 2:3

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Volume of a sphere with radius 3 (in terms of π):

a. 72π

b. 36π

c. 18π

d. 47π

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A cone has radius 3 and height 4. Find the slant height.

a. 7

b. 5

c. 8

d. 6

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Area of a triangle with base 8 and height 5:

a. 13

b. 40

c. 22

d. 20

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Surface area of a sphere with radius 3 (in terms of π):

a. 72π

b. 49π

c. 36π

d. 18π

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two similar spherical storage tanks: Tank B has a surface area 9 times that of Tank A. If Tank A holds 250 liters, how many liters does Tank B hold?

a. 2,250 liters

b. 6,750 liters

c. 1,500 liters

d. 5,625 liters

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A topographical map has a scale of $1:25{,}000$. A regional forest reserve has an area of $8\ \text{cm}^2$ on the map. What is the actual ground area of the reserve in square kilometers (km^2)?

a. 0.5 km²

b. 2.0 km²

c. 5.0 km²

d. 0.2 km²

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

If each dimension of a rectangle is increased by 40%, by what percentage does the area of the rectangle increase?

a. 96%

b. 40%

c. 80%

d. 140%

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A rectangle with length 4 and width 13 has a semicircle of diameter 13 attached to one of the short sides. What is the total area of the composite figure, rounded to two decimal places?

a. 118.37

b. 123.37

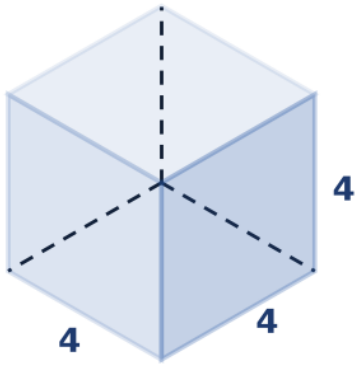
c. 184.73

d. 52

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Circles

What is the surface area of a cube with edge length 4 cm?



a. 64 cm²

b. 96 cm²

c. 128 cm²

d. 144 cm²

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

A trapezoid has area 48 and height 6. Find the sum of its two parallel sides.

a. 8

b. 6

c. 18

d. 16

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Foundations & Reasoning

A kite has diagonals 9 and 11. Find its area.

a. 40

b. $\frac{105}{2}$

c. $\frac{99}{2}$

d. 99

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Coordinate Geometry

Area of triangle base 11 height 4:

a. 15

b. 22

c. 44

d. 25

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability

A rhombus has diagonals 10 and 12. Find its area.

a. 22

b. 60

c. 62

d. 120

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Lines & Angles

Triangle with sides 13, 14, 15. Find its area using Heron's formula.

a. 86

b. 81

c. 84

d. 91

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